Bus Management System

[From the Haramain high speed railway to Al-Haram]
[Group 14]

Project Team

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CS 1312: Computer Programming 2
Instructor: Dr. Hana Alghamdi

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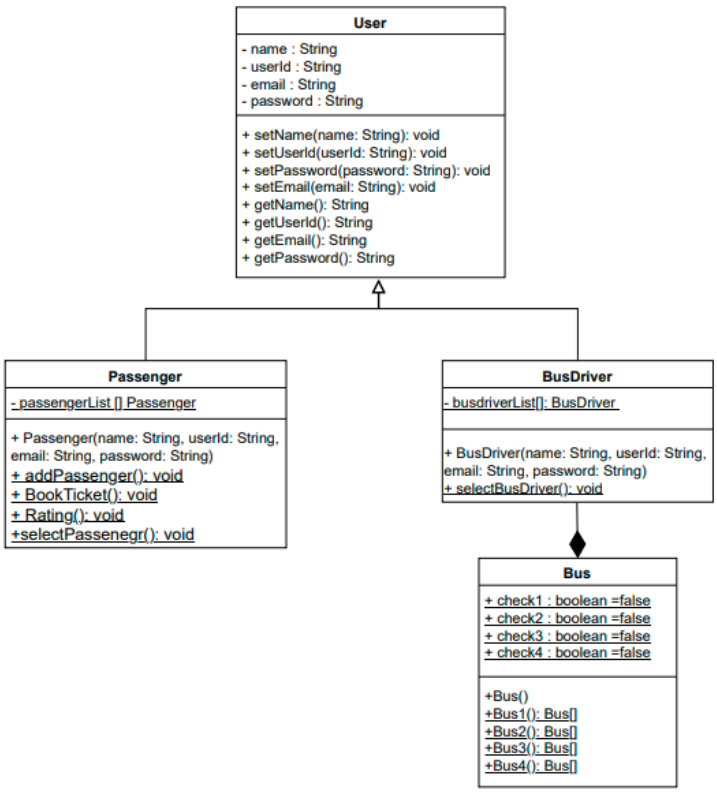
1- Project Title

Bus Management System

2- Project Description

Our Java project will be on a bus management system. The application will perform the main task, which are: collecting information from the passengers, drivers, scheduling trips From the Haramain high speed railway to Al-Haram, and taking feedbacks about the quality of the trip from users.

3- Project UML



4- Project Functionalities or Tasks

Our project is an application for al-Haram bus management that serves (the passengers and the bus drivers) to book trips at certain hours and deliver the passenger as soon as possible. There are methods for every operation such as addPassenger that add a new passenger to the list. BookTicket is a method that create a new booking and departure time for passengers. Rating is a method to receive feedbacks and ratings from passengers. selectPassenegr is a method to let the passenger choose a service. SelectBusDriver is a method that takes the driver's info.

5- Project workflow

The execution starts with asking the user to choose if he is a passenger or a bus driver, or to enter - 1 to exit. Based on what the user chose, a list of available services will drop down.

if the user chose bus driver the program is going to ask for his full information so that if there's a booking in his schedule, he will be informed of all the details about the trip, if not, the program respectfully inform him that at this moment his service is not needed. And if the user chose passenger the following is displayed:

1-add a new passenger
2-book a ticket
3-rating
4-go back to main page.

*Choosing No.1 will lead to asking the passenger for his information and then inform him that he's added to the system successfully.

*Choosing No.2 will lead to asking that passenger to select a departure time (6:00am,8:00am, 10:00, 12:00pm) and then inform him of the bus no that will transport him, and to be on time so that he don't miss the bus.

*Choosing No.3 will lead to prompt the passenger to enter his feedback about the trip and to rate it out of 5.

*Choosing no4 will lead to get the passenger back to the main page.

6- The distribution of the work

This project is all teamwork. We've been meeting and opening zoom calls to write this program together and share every idea. so, there's no specific task for each member. We all wrote this program together and we all contribute equally to make this program work.

7- Lessons Learned

* We developed our teamwork skills.
* We learned how to create a program to organize bus trips and how to book bus tickets
* We learned how to make UML well in a professional way.
* We learned how to deal with the errors and how to make it correct as we want without error in the program.

8- Project difficulties

* The biggest difficulty was finding a time that suits everyone.
* We did a whole different class and UML before we got to this final project.
* We changed the UML, so many times.
* At first, we didn’t know how to connect the classes together and still make the main class short and clear.
* We wrote a lot of classes and then realized that we could merge them into fewer classes in a better way.
* We were missing some information and we were lost at the beginning, but it was clarified to us in the lectures that we took later.

9- Project Declaration

☑ ☐ We confirm that the work of this project was solely undertaken by ourselves and that no help was provided from other sources as those allowed. As well as we confirm that we completely aware of the violation consequences of the academic integrity.

Read and check the previous declaration.

List all your IDs:

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